

Experiment-4

1. Mistakes in the Original Algorithm:

The given algorithm works only when the second number (`num2`) is positive. There are a few problems with it:

- It directly uses `num2` as the number of repetitions. So if `num2` is negative, the loop does not work correctly.
- It does not consider whether the final answer should be positive or negative.
- Zero is not handled separately.
- It always uses `num2` as the loop counter. This does **not guarantee minimum additions**. For example, for 3×12 , adding 3 twelve times is unnecessary; we can add 12 only three times.
- Therefore, we should repeat the addition using the number with the **smaller absolute value**.

2. Corrected Algorithm:

The basic idea is simple: first deal with zero and signs, then repeatedly add the larger number only as many times as the smaller number requires.

1. Start
2. Read `num1`, `num2`
3. If `num1 = 0` OR `num2 = 0`
 Print 0
 Stop
4. Set `sign = 1`

5. If $\text{num1} < 0$
 $\text{num1} = -\text{num1}$
 $\text{sign} = -\text{sign}$
6. If $\text{num2} < 0$
 $\text{num2} = -\text{num2}$
 $\text{sign} = -\text{sign}$
7. If $\text{num1} < \text{num2}$
 Swap num1 and num2
8. Set $\text{prod} = 0$
9. Set $\text{count} = 1$
10. While $\text{count} \leq \text{num2}$
 $\text{prod} = \text{prod} + \text{num1}$
 $\text{count} = \text{count} + 1$
11. If $\text{sign} = -1$
 $\text{prod} = -\text{prod}$
12. Print prod
13. Stop

> *Why does this use minimum additions?*

After converting both numbers to positive values, we make sure that **num1** is the larger number and **num2** is the smaller one.

So:

$\text{prod} = \text{prod} + \text{num1}$

is performed only **num2** times.

Therefore:

Number of additions = min(|original num1|, |original num2|)

For Example:

$$12 \times 3$$

$$0 + 12 = 12$$

$$12 + 12 = 24$$

$$24 + 12 = 36$$

Only **3 additions** are needed.

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